SUPPLEMENTAL MATERIAL | IEEE SUBMISSION CONTEXT

Strategic Case for Project TIWALA

Contextual Background & Implementation Scenario for "Orchestrating Agricultural Intelligence for Smallholder Rice Farmers"

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1 Executive Summary

Project TIWALA is an advanced socio-technical framework designed to resolve the **Data Fragmentation Challenge** in Philippine agriculture. By orchestrating fragmented data sources from national agencies into a Unified Data Mesh, it delivers hyper-personalized, climate-resilient intelligence directly to smallholder farmers. This document outlines the strategic rationale for selecting an Enterprise-Grade architecture to meet urgent national deployment needs.

DISCLAIMER

*Project TIWALA is an independent research initiative (NU-Manila, Q4 2025). While this paper proposes a government-scale architecture, TIWALA is a registered non-profit NGO and is **not** currently affiliated with the Department of Agriculture.*

2 User Profile & Dilemma

ATTRIBUTE	DETAIL
Demographic	Age 58 (Avg); 52% High School Education
Tech Access	96% Smartphone ownership, but low digital literacy
Core Dilemma	**Information Asymmetry:** High production costs vs. low farmgate prices. Vulnerable to climate shocks.
Trust Factor	Skeptical of "unsure if true" data; relies on peers.

3 The Business Case

Scenario: "National Rapid Response Agri-Intelligence Initiative"

To justify the Enterprise-Grade architecture proposed in the IEEE paper, we present a realistic procurement scenario:

THE CLIENT
Special Task Force (DA)

BUDGET
RCEF/Deep Pockets

TIMELINE
Critical Urgency (Pre-Typhoon Season Launch)

Target Data Sources

DA/PhilRice	Unstructured Research (PDFs)
PAGASA	Real-time Weather APIs
BSWM	Geospatial Soil Maps (WMS)
LBP	Financial/Credit Data

Architectural Rationale

WHY MICROSOFT FABRIC?

Speed over Cost

SaaS model enables rapid data ingestion, meeting the urgent 6-month deadline vs. 18 months for custom build.

WHY VIRTUALIZATION?

Political Strategy

"Zero-Copy" via Shortcuts lets agencies keep ownership while granting TIWALA real-time access.

WHY GENERATIVE AI?

Usability Gap

Azure OpenAI (RAG) translates complex data into vernacular advice, ensuring a No-Learning-Curve Interface.

TECH STACK COMPONENTS